TECTONICS, VOL. 13, NO. 4, PAGES 917-928, AUGUST 1994

Paleomagnetism of baked sedimentary rocks in the Newark and Culpeper basins: Evidence for the J1 cusp and significant Late Triassic apparent polar wander from the Mesozoic basins of North America

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Abstract. A paleomagnetic study of 14 sites in the baked sedimentary rocks of the Newark basin Passaic Formation in southeastern Pennsylvania reveals two types of magnetic behavior. Dark gray-colored, baked sedimentary rocks have peak unblocking temperatures of 640°C, high magnetic intensities, and shallow, normal polarity, northeasterly directions. Light gray-colored rocks have peak unblocking temperatures of less than 580°C, low magnetic intensities, and intermediate inclination, normal polarity, northwesterly directions. The low unblocking temperature magnetizations are secondary magnetizations which have declinations similar to but are shallower than the B remagnetization observed by Witte and Kent (1991) throughout the Newark basin. The discrepancy may be due to "underprinting" by an unresolved primary magnetization. The low unblocking temperature magnetization was probably acquired by growth of secondary magnetite during a hydrothermal event, as postulated by Sutter (1988), based on geochronologic data. The high unblocking temperature magnetization is significantly prefolding. Both the low-peak unblocking temperature magnetization and the high-peak unblocking temperature magnetization suggest a 15° counterclockwise block rotation of the Sassamansville syncline. If this rotation is removed from the high unblocking temperature sites collected around the fold, a stronger passage of the fold test results. Six sites were also collected from baked sediments and one site from diabase in northern Virginia's Culpeper basin, since Sutter's geochronological work indicated that the intrusives in the Culpeper basin are coeval to the Newark basin intrusives. Virtual geomagnetic poles (VGPs), based on the tilt-corrected, high-temperature Newark basin magnetizations, were compared with the VGPs calculated from the site means of a high-temperature magnetization isolated from baked sedimentary rocks in the Culpeper basin and to the magnetizations reported by Raymond (1982) from dikes and sills. In this comparison the in situ Culpeper poles agreed with the prefolding Newark poles significantly better than the prefolding Culpeper poles. This result indicates that Culpeper intrusives erupted into already tilted sedimentary rocks. The paleomagnetic pole determined from the combined Culpeper baked sediments, dikes, and sills (in situ coordinates) and the Newark basin baked sediments (tilt-corrected coordinates) lies at 60°N,69°E and is of 201 Ma age. This latest Triassic/earliest Jurassic pole, when combined with the Newark basin Carnian results (Witte and Kent, 1989) and Norian results (Witte et al., 1991) corrected for a counterclockwise block rotation (Kodama et al., 1994), provides a record of significant polar wander from eastern North America's Mesozoic basins for the Late Triassic. This is consistent with observations made for a similar time period from rocks on the Colorado Plateau (Bazard and Butler, 1991). Comparison of the Newark/Culpeper pole to similar age poles from the Kayenta (Bazard and Butler, 1991) and Moenave Formations (Ekstrand and Butler, 1989) only requires small amounts (5°) of Colorado Plateau rotation. The pole also provides the first well-dated evidence of the J1 cusp in North American apparent polar wander from rocks not located on the Colorado Plateau, thus giving strong support for the usefulness of paleomagnetic Euler pole analysis of apparent polar wander.

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Paper number 94TC00294.
0278-7407/94/94TC-00294$10.00

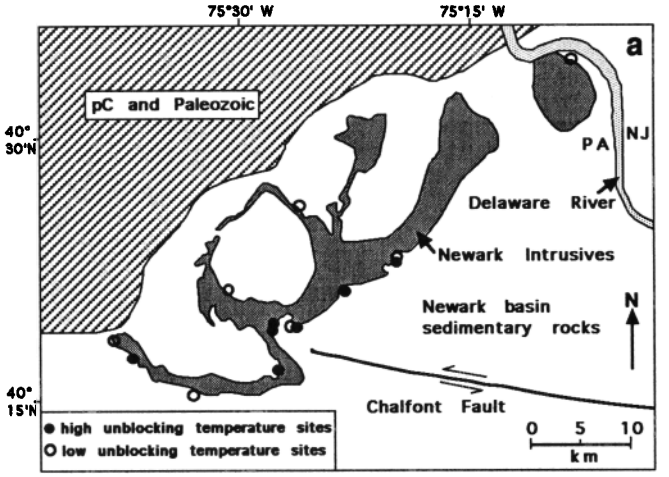
Introduction

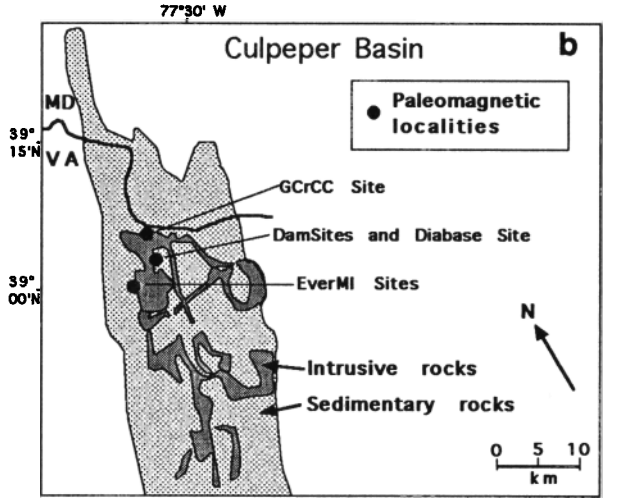
The Late Triassic/Early Jurassic portion of North America's apparent polar wander path is currently the focus of intense study by paleomagnetists. *Witte and Kent* [1989, 1990], *Kent and Witte* [1993] and *Witte et al.* [1991] have conducted a detailed paleomagnetic reexamination of the red sedimentary rocks in the Newark basin which range in age from latest Carnian to Hettangian. Their results indicate that North America has not experienced significant apparent polar wander for approximately 20 m.y. and therefore suggest a "stillstand" for North America during the latest Triassic/earliest Jurassic. *Bazard and Butler* [1991] have conducted a detailed paleomagnetic restudy of the Chinle and Kayenta Formations, located on the Colorado Plateau in the western United States, which provides an entirely different picture of apparent polar wander for the 30-m.y. period from the Norian to the Pliensbachian. Bazard and Butler observe significant apparent polar wander during this time period, including a sharp cusp in the apparent polar wander path during the Sinemurian when the sense of apparent polar wander reverses direction.

The resolution of the differences between these two different pole paths has important implications for the regional tectonics of the western United States and the global plate tectonics of North America. In order to explain the discrepancy between the Newark basin data and the Chinle/Kayenta results,

Kent and Witte [1993] have proposed that the Colorado Plateau has experienced a significant clockwise rotation of greater than 10°. This result is in direct contradiction to *Bryan and Gordon's* [1986, 1990] examination of a much larger data set of paleopoles on and off the plateau that range in age from Paleozoic to Mesozoic and indicate less than 5° of clockwise rotation. By suggesting that Colorado Plateau rotation might be as large as 10° to 15°, *Kent and Witte* [1993] also bring into question the reality of the sharp cusp in the North American apparent polar wander that is postulated to occur near the Triassic-Jurassic boundary [*Gordon et al.*, 1984; *May and Butler*, 1986] and has been denoted the J1 cusp. The evidence for the cusp comes primarily from data collected on the Colorado Plateau, so large clockwise plateau rotations might reduce or even remove any basis for the cusp. *Smith* [1987] has reported paleopoles that lie near the J1 cusp from North Carolina diabase dikes, but their age is not well constrained. The existence of the cusp, however, is important to the acceptance of paleomagnetic Euler pole (PEP) analysis of apparent polar wander. In PEP analysis a model of plate motion is used to construct apparent polar wander paths in which lithospheric plates are assumed to rotate around fixed Euler poles for geologically significant time periods (tens of millions of years). In the PEP approach, long, small-circle tracks of constant direction apparent polar wander should be separated by sharp cusps, which indicate times when plate-driving mechanisms and hence the direction of plate motion and apparent polar wander changed. This would be represented in the apparent polar wander path by the establishment of a new Euler pole. These time periods might correspond to significant geological events at plate boundaries, so that the existence of a cusp in the apparent polar wander path should lead geologists to examine the geologic record for these events [*Gordon et al.*, 1984]. Evidence of high-latitude paleopoles in the Middle Jurassic [*Van Fossen and Kent*, 1990; *Witte and Kent*, 1991; *Van der Voo*, 1992; *Kodama et al.*, 1994; *Kodama and Mowery*, 1994] has already brought into question the long track proposed by PEP analysis for the Jurassic. The demise of the J1 cusp would further erode the evidence for the applicability of PEP analysis and change our conception of plate dynamics. In addition to affecting our thinking about plate driving mechanisms, the inclusion of the J1 cusp in North American apparent polar wander creates a different paleogeography for North America during the breakup of Pangea in the latest Triassic-earliest Jurassic than proposed by *Ziegler et al.* [1983]. North America, based on the J1 cusp, would be at lower latitudes and moving faster than originally envisioned [*Bazard and Butler*, 1991].

Paleomagnetic studies of the sedimentary and igneous rocks folded at the Sassamansville syncline along the border fault of the Newark basin in southeastern Pennsylvania [*Kodama et al.*, 1994; *Kodama and Mowery*, 1994] provide the basis for the





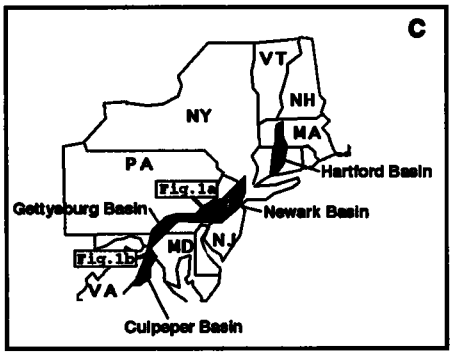


Figure 1. (a) Location map shows paleomagnetic sites in the southeastern Newark basin, including high unblocking temperature sites (solid circles) and low unblocking temperature sites (open circles). Gray shaded areas are diabase intrusive rocks. Striped pattern indicates undifferentiated Precambrian (pC) and Paleozoic rocks. Geology is based on *Lyttle and Epstein's* [1987] 1°x2° Newark Quadrangle. (b) Location map for paleomagnetic sites in the northern Culpeper basin. Note that three sites (EverMI5, 6, and 7) were collected at the Evergreen Mills locality, and two baked sediment sites (DamSites2 and 3) and one diabase site (DiabDS) were collected near the Goose Creek Dam. The Goose Creek Country Club (GCrCC) site was collected across the road from the entrance to the Goose Creek Country Club. Dark gray shading indicates intrusive igneous rocks; light gray, Mesozoic sedimentary rocks. Geology is based on *Gore et al.*'s [1989] map of the Culpeper basin. (c) Index map is for Figures 1a and 1b.

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work reported here. Both the prefolding characteristic remanence and the synfolding secondary magnetization of the Passaic Formation exposed at Sassamansville indicate a 15° counterclockwise block rotation of the region. The diabase which intrudes the Passaic Formation has a synfolding magnetization which also supports a block rotation, but the failure of a baked contact test indicates that the diabase has been totally remagnetized at the same time that the sedimentary rocks acquired their secondary magnetization. A hydrothermal event postulated by *Sutter* [1988], based on geochronological data for the Newark basin, is the assumed cause of the remagnetization of the Newark basin sedimentary [*Witte and Kent*, 1991] and igneous [*Kodama and Mowery*, 1994] rocks. The baked contact test conducted as part of the diabase paleomagnetic study [*Kodama and Mowery*, 1994] also revealed a baked sediment magnetization at one site that produced a virtual geomagnetic pole very close to the J1 cusp. Since *Sutter*'s [1988] geochronologic work on the intrusive rocks of the Newark and Culpeper basins suggests a crystallization age for the diabase of 201 Ma, which is close to the postulated age of the J1 cusp, a fold test was conducted on the baked sediments in the Newark basin in order to determine if they carry a primary crystallization age magnetization, even though the diabasic rocks that caused their thermal resetting have been remagnetized [*Kodama and Mowery*, 1994]. Since *Sutter*'s [1988] $^{40}Ar/^{39}Ar$ work on Mesozoic basin intrusive rocks also indicates that the Culpeper basin intrusives are indistinguishable in age from the Newark basin intrusives, baked sediments and one diabase site were also sampled in the northern Culpeper basin in northern Virginia.

Methods

Oriented 25-mm-diameter cores were drilled at 14 sites in the Passaic Formation, where it had been baked by the diabasic rocks of the Sassamansville syncline and by diabase which outcrops as a large ring dike to the north and east of the Sassamansville syncline (Figures 1a and 1c). Four to 11 independently oriented cores were collected from each site in the Newark basin in southeastern Pennsylvania. In addition, six sites were sampled in the baked sediments associated with the intrusive rocks of the northern Culpeper basin in Virginia (Figures 1b and 1c). Two of these sites were collected near a dam across Goose Creek. A nearby (<100 m) exposure of diabase was also sampled to conduct a partial baked contact test. Unfortunately, no unbaked sediments outcropped within several kilometers, so a complete baked contact test was not possible. Five to 11 independently oriented cores were collected at each of the Culpeper basin sites. All cores sampled at the Newark and Culpeper basins were drilled with a diamond-coring drill and oriented with magnetic and sun compasses. Detailed, progressive thermal demagnetization in at least six steps (10 or more steps for the hematite-bearing samples, see Results section) was used to isolate the characteristic remanence (ChRM) for each sample. Principal component analysis [*Kirschvink*, 1980] was used to calculate each sample's ChRM. Site mean directions and their associated precision parameters and confidence intervals were determined using *Fisher* 's [1953] statistics. An incremental fold test was conducted for the Newark basin sites, using the bedding attitude measured for the baked sediment sites. The statistical significance of the mean of the site means with the greatest precision parameter K, [*Fisher*, 1953], or the minimum f statistic, [*McFadden and Jones*, 1981] was tested using the techniques of either *McElhinny* [1964] or *McFadden and Jones* [1981], respectively. Although McElhinny's statistical test has been shown to be incorrect [*McFadden and Jones*, 1981], it is generally more demanding than the McFadden and Jones test, even though the McFadden and Jones approach is the most commonly used technique for testing the significance of the fold test. It has been used, rather than the more recently formulated *McFadden* [1990] test, to allow easy comparison of our results with previous paleomagnetic studies. The McElhinny technique will be used in this paper to demonstrate which magnetizations pass the most stringent of fold tests.

Results

Newark Basin

Thermal demagnetization of the baked sediments sampled from the Newark basin revealed two distinct types of behavior. Sites with high-intensity natural remanent magnetizations (NRMs) (1-5 A/m) showed linear decay into the origin on orthogonal vector endpoint diagrams [*Zijderveld*, 1967], with peak unblocking temperatures typically as high as 645°C (Figure 2). Sites with lower NRMs (0.05-0.1 A/m) contained samples also showing linear trends into the origin during demagnetization and peak unblocking temperatures between 560°C and 580°C (Figure 2). The directions associated with the different demagnetization behavior were also distinct. High unblocking temperature sites showed normal polarity and shallow inclination directions with northeasterly declinations (Table 1). Low unblocking temperature sites had north-northwesterly declination, intermediate inclination, and normal polarity directions (Table 2). The difference in magnetic behavior was reflected in the color of the baked sediments. The high-intensity NRM, higher unblocking temperature samples were dark gray in color. The low-intensity, lower unblocking temperature samples were light gray in color. Seven of the 14 sites collected from the Newark basin baked sediments showed the high unblocking temperature behavior; four sites were located around the Sassamansville syncline and three were located south of the large ring dike to the east of the syncline. Six of the 14 sites exhibited low unblocking temperature behavior, three from around the Sassamansville fold and three from sites north and east of the fold. Only one site (Pruss Hill dam site) collected from what were mapped to be baked sediments [*Longwill and Wood*, 1965] on the southern limb of the Sassamansville fold, had the high unblocking temperature (680°C) low negative inclination, and reversed polarity directions typical of unbaked Passaic Formation sedimentary rocks [*Kodama et al.*, 1994].

When the site mean directions for the high unblocking temperature sites are unfolded in an incremental application of the fold test [*Graham*, 1949] (Figure 3), the precision parameter [*Fisher*, 1953] shows a peak at 100% unfolding. The *McFadden and Jones* [1981] fold test statistic f shows a minimum at 100% unfolding. Both results indicate the best clustering of site means at 100% unfolding. Although

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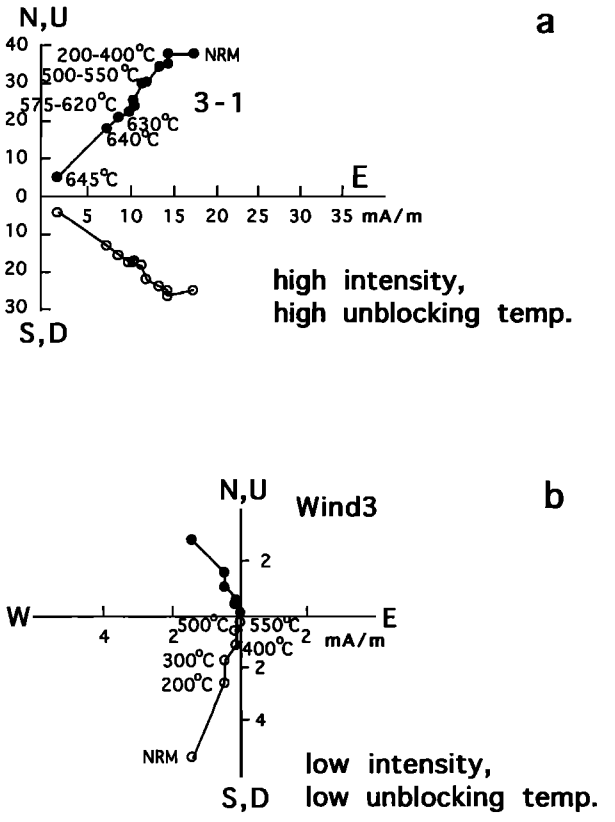


Figure 2. Orthogonal vector endpoint diagrams [*Zijderveld*, 1967] are shown for thermal demagnetization of two samples from Newark basin baked sediments. These two samples represent the two different types of demagnetization behavior observed for these rocks. (a) High unblocking temperature rocks typically have peak unblocking temperatures of 645°C, univectorial decay into the origin, and high-intensity natural remanent magnetizations (NRMs). (b) Low unblocking temperature rocks have peak unblocking temperatures of 550° to 560°C, univectorial trend into the origin, and low-intensity NRMs. Open circles indicate projection on a vertical plane; solid circles indicate projection on the horizontal.

McElhinny's [1964] significance test does not indicate that the remanence is a statistically significant prefolding remanence, McFadden and Jones' statistical test does indicate that it is prefolding at the 95% confidence level. At 100% unfolding the site means appear scattered along a small circle, coaxial to a vertical axis. Since previous work in this area indicates that the Sassamansville fold has apparently been rotated 15° counterclockwise with respect to other parts of the Newark basin [*Kodama et al.*,1994; *Kodama and Mowery*, 1994], the incremental fold test was applied separately to the four high unblocking temperature sites on the fold and the three sites off the fold. In both cases the magnetization showed peak clustering at 100% unfolding; in the case of the four sites on the fold this prefolding configuration was strongly significant (95% confidence level [*McElhinny*, 1964]). The three sites off the fold had very similar northwesterly dipping bedding attitudes, so the increase in the precision parameter was not significant at the 95% confidence level. When the mean of the four site means on the fold is compared with the mean for the sites off the fold in stratigraphic coordinates, the baked sediment magnetizations indicate a 20.5° (+/- 8.4°) counterclockwise rotation for the sites on the fold and insignificant latitudinal offset (0.64°+/- 8.4°) [*Demarest*, 1983]. This result is consistent with the counterclockwise rotation of the fold observed for the sedimentary [*Kodama et al.*, 1994] and igneous [*Kodama and Mowery*, 1994] paleomagnetic results. If a 15° counterclockwise rotation is removed from the mean directions and bedding strike of the four sites located on the fold and an incremental fold test is conducted using all seven sites, both K_{max} and f_{min} occur at 100% unfolding (Figure 4). However, in this case the K_{max} mean direction is significantly prefolding at the 95% confidence level using the *McElhinny* [1964] significance test, and there is no indication of the site means being scattered along a small circle.

The plunge of the fold axis, determined from the bedding attitude of the four sites on the fold (295° trend, 11° plunge), has little affect on the K_{max} mean direction for the four sites on the fold. The results reported above are based on unfolding around local strike. If the fold test is conducted by first unplunging site mean directions and site bedding attitudes and then unfolding the site means, then the K_{max} direction changes by less than 1°. If the site mean directions are unfolded around the fold axis but not unplunged, then the K_{max} direction changes by 3.2°.

When an incremental fold test is conducted using the site mean directions for the low unblocking temperature sites, the precision parameter peaks at 75% unfolding, although this clustering is statistically insignificant [*McElhinny*, 1964]. It was not possible to conduct the *McFadden and Jones* [1981] test, since only one site was located on one of the two limbs of the fold. When the sites are split into those on the fold (N=3) and those off the fold (N=3) and the mean directions are com-

Table 1. High Unblocking Temperature Newark Basin Baked Sediment Site Means

Site	Latitude, °N	Longitude, °E	N/N_0	Incl.	Decl.	K	α_{95}	Dip	Dip Dir
Rte 29N	40.356	284.517	5/5	5.0	13.7	157.1	5.0	20	242
Rte 29Na	40.356	284.517	6/6	-16.3	3.9	158.6	4.5	20	242
Rte 100	40.290	284.358	4/4	17.2	8.7	517.7	3.1	13	45
Rte 29S	40.275	284.524	5/5	33.1	20.6	728.6	2.6	30	335
Rte 63rv	40.296	284.558	11/11	27.0	50.9	168.0	3.3	36	340
PA Turn	40.349	284.609	7/7	5.6	30.8	224.8	3.5	9	352
Rte 309rv	40.386	284.671	4/6	28.7	26.2	1074.2	2.1	16	345

First four sites located on Sassamansville syncline. N/N_0, number of samples used/number of samples collected. Incl., inclination in degrees; Decl., declination in degrees; K, precision parameter; α_{95}, radius of 95% confidence about the mean; Dip, degrees; Dip Dir, direction in degrees.

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Table 2. Low Unblocking Temperature Newark Basin Baked Sediment Site Means

Site	Latitude, °N	Longitude, °E	N/N_0	Incl.	Decl.	K	α_{95}	Dip	Dip Dir
Rte 663N	40.420	284.527	4/6	41.6	2.6	622.9	2.8	15	350
Rte 663S	40.352	284.459	5/5	30.6	352.9	91.4	6.6	7	20
PrussHill	40.262	284.428	6/6	36.0	331.0	42.2	8.8	18	345
Winding	40.304	284.341	6/9	44.6	343.0	35.7	9.6	24	23
Ring Rx	40.561	284.877	3/6	36.0	343.0	130	7.1	14	354
Rte 309	40.386	284.671	-	-	-	-	-	-	-
Rte 63	40.296	284.558	3/10	49.4	9.2	62.8	10.2	36	340

First four sites located on Sassamansville syncline. Abbreviations are the same as for Table 1.

pared at 50% unfolding, a 10.6° (+/-12.1°) counterclockwise rotation for the sites on the fold is indicated. Finally, when the 15° counterclockwise rotation is removed from the sites on the fold and the incremental fold test is repeated, the precision parameter peaks at 60% unfolding, but it is again statistically insignificant (Figure 5) [*McElhinny*, 1964]. The K_{max} direction (declination(dec), 357.7°; inclination(inc), 29.4°; α_{95}(radius of 95% confidence about the mean), 6.4°) has a declination similar to *Witte and Kent*'s [1991] B remagnetization for the Newark basin but an inclination approximately 13° shallower.

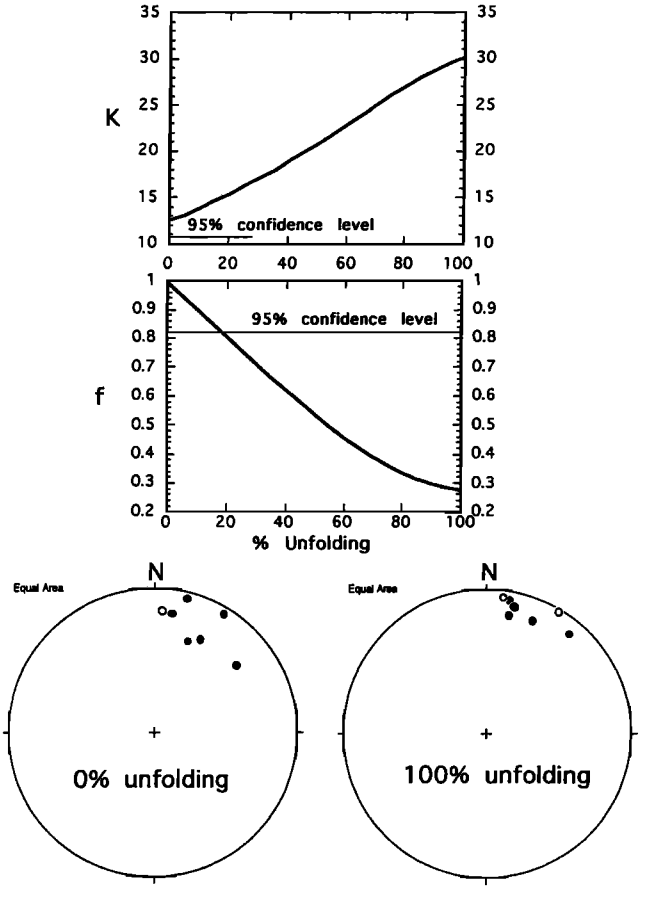


Figure 3. Incremental fold test results are for seven high unblocking temperature sites collected from the Newark basin in southeastern Pennsylvania. (top) Graphs show the variation of the precision parameter K and *McFadden and Jones'* [1981] fold test statistic f during incremental unfolding. The magnetization is significantly prefolding at the 95% confidence level, using the f statistic [*McFadden and Jones*, 1981] but not K [*McElhinny*, 1964]. (bottom) Stereonets are equal area nets and show site means in geographic (0% unfolding) and stratigraphic (100% unfolding) coordinates. Open circles are upper hemisphere; solid circles, lower hemisphere. Note the scatter of the 100% unfolding site means along a small circle concentric to a subvertical axis.

Culpeper Basin

The two types of magnetic behavior observed in the Newark basin baked sediments were not observed in the Culpeper basin baked sediments. At the six sites sampled, thermal demagnetization resulted in univectorial decay into the origin on vector endpoint plots [*Zijderveld*, 1967] to peak unblocking temperatures at least as great as 560°C (the two dam sites; Table 3), but typically 650° to 680°C (all three EverMls sites; Table 3 and Figure 6). This is comparable with the behavior seen in the high unblocking temperature sites in the Newark basin. Directions were all normal polarity with north-northeasterly declinations and shallow to intermediate inclinations. The site mean from the diabase is statistically indistinguishable from the site mean of the closest baked contact site (DamSite3; Table 3) at the 95% confidence level [*McFadden and Lowes*, 1981]. However, because of poor demagnetization behavior at DamSite3 this site mean is only based on two samples. This result, and the inability to find any nearby unbaked sediment to sample, yields a weak passage of the baked contact test.

Since all the Culpeper basin sites dip toward the west, application of an incremental fold test shows only marginally better clustering of the baked sediment site means at 80% unfolding. This result is not statistically distinguishable from either the in situ or completely unfolded clustering of the data.

Discussion

Newark Basin

The results from the high unblocking temperature sites in the Newark basin indicate that the characteristic remanence of these sites was acquired before tilting of these beds. The magnetization is probably a complete thermal overprint caused by the intrusion of the Sassamansville diabase at 201 Ma [*Sutter*, 1988]. However, the statistically significant difference between the high unblocking temperature baked sediment mean direction (dec, 18.8°; inc, 8.1°; α_{95}, 9.6°) in stratigraphic coordinates and the Sassamansville diabase mean direction (inc, 42.1°; dec, 349.7°; α_{95}, 5.9°), combined with the strong similarity of the synfolding diabase direction to the

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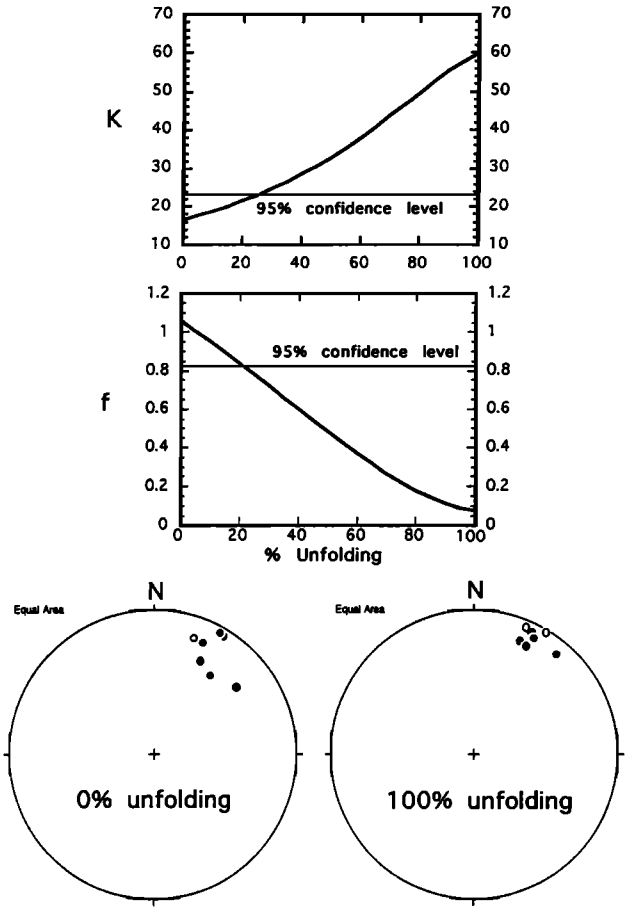


Figure 4. Incremental fold test results are for seven high unblocking temperature sites from the Newark basin, but with a 15° counterclockwise rotation removed from the four sites located in the Sassamansville syncline, which previous studies [*Kodama et al.*, 1994; *Kodama and Mowery*, 1994] indicate has been rotated with respect to the rest of the Newark basin. Note that the 100% unfolding site means do not show the scatter observed in Figure 3 and that the magnetization is significantly prefolding at the 95% confidence level for both the *McFadden and Jones* [1981] and the *McElhinny* [1964] fold tests. Symbols are the same as for Figure 3.

B component remagnetization direction observed by *Witte and Kent* [1991] throughout the Newark basin, provides strong evidence that the diabase has been remagnetized, subsequent to its crystallization [*Kodama and Mowery*, 1994]. Apparently, the high unblocking temperature baked sediment sites have not been affected by this remagnetization event. In addition, the high unblocking temperatures of these sites are consistent with a thermal remagnetization of the hematite-bearing red sediments of the Passaic Formation. Comparison of the high-temperature site means on and off the Sassamansville syncline also provides further evidence for a 15° counterclockwise rotation of the area immediately around the Sassamansville fold. The proximity of the sites off the fold helps to better delineate the area affected by the block rotation and strongly points to movement on the intrabasinal Chalfont fault as the cause of the block rotation (Figure 1a). *Schlische* [1992] argues from geological observations that left-lateral motion may have occurred on the Chalfont fault, and *Kodama et al.* [1994] and *Kodama and Mowery* [1994] suggest that this motion could be one possible cause of the 15° counterclockwise rotation of the Sassamansville fold. Given the strong evidence for a block rotation from the sedimentary rocks [*Kodama et al.*, 1994], diabase [*Kodama and Mowery*, 1994], and baked sediments and the improvement in the statistical significance of the prefolding mean direction when the rotation is removed from the four sites on the fold, all subsequent analyses will be based on results in which a 15° counterclockwise rotation has been removed from the relevant sites.

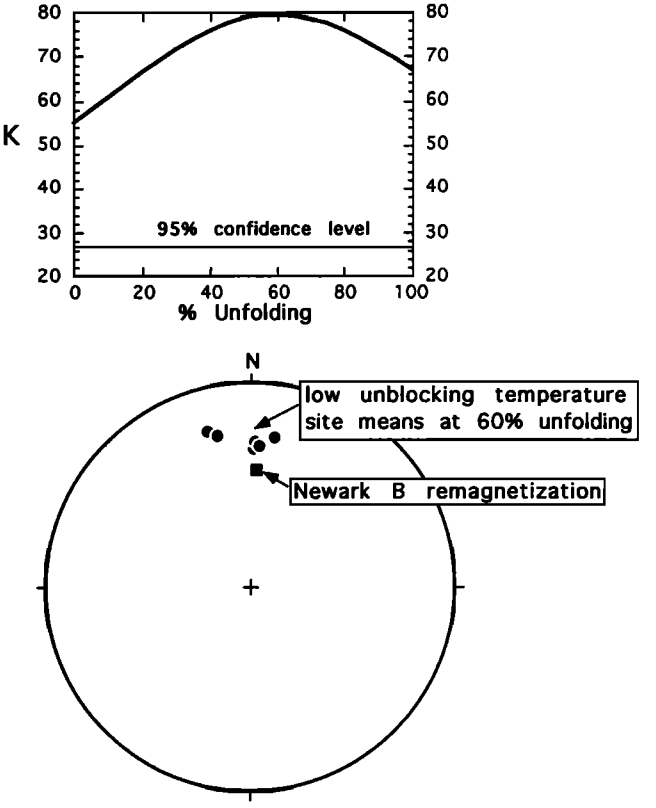


Figure 5. Incremental fold test results are for the six low unblocking temperature sites from the Newark basin with a 15° counterclockwise rotation removed from the three sites located on the Sassamansville fold. Note that the site means cluster best at 60% unfolding, similar to what was found for the secondary magnetization of the Sassamansville diabase [*Kodama and Mowery*, 1994]. However, the low unblocking temperature magnetization is not significantly synfolding at the 95% confidence level [*McElhinny*, 1964]. The Newark B remagnetization mean direction has been isolated by *Witte and Kent* [1991] from other parts of the Newark basin and has been interpreted to be a secondary magnetization caused by a hydrothermal event at about 180 Ma. The low-temperature site means are similar to the Newark B remagnetization direction but shallower by about 13°, possibly due to underprinting by the high unblocking temperature direction. Symbols are the same as for Figure 3.

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Table 3. Culpeper Basin Baked Sediment and Diabase Site Means

Site	Latitude, °N	Longitude, °E	N/N_0	Incl.	Decl.	K	α_{95}	Dip	Dip Dir
DamSite2	39.057	282.475	5/6	28.6	8.5	133.0	5.4	36	269
DamSite3	39.057	282.475	2/5	38.9	14.0	1485.0	2.6	33	268
DiabDS	39.057	282.475	5/5	36.7	14.6	597.4	2.6	-	-
GCrCC	39.087	282.486	11/11	27.8	3.1	41.5	3.1	33	241
EverMl5	39.017	282.424	7/7	27.2	19.5	43.9	8.0	38	268
EverMl6	39.017	282.424	7/7	20.1	23.0	152.9	4.3	40	281
EverMl7	39.017	282.424	9/9	20.0	12.5	60.4	6.0	37	275

Abbreviations same as for Table 1.

The low unblocking temperature results are magnetically and directionally distinct from the high unblocking temperature results and are much closer to the B remagnetization direction observed by *Witte and Kent* [1991] for the Newark basin. The lack of hematite, the magnetic mineral in the Passaic Formation which should be thermally reset by the intrusion of the diabase, in the low-temperature sites and the presence of magnetite, based on the peak unblocking temperature of 560°C, suggest that the primary magnetic minerals in the low-temperature sites have been removed by the hydrothermal fluids postulated by *Sutter* [1988] for the Newark basin and have been replaced by secondary magnetite deposited by those fluids. This would be the same mechanism proposed by *Kodama and Mowery* [1994] for the remagnetization of the diabase and is based on petrographic observation of secondary magnetite in the diabase [*Davidson and Wyllie*, 1968]. Apparently, variation in hydrologic parameters (porosity and permeability) dictated which parts of the baked sediments would be overprinted by the hydrothermal fluids and which parts would not be. The observation of dark and light gray "striping" of the baked sediments subparallel to bedding along the Northeast Extension of the Pennsylvania Turnpike near the intersection with Pennsylvania Route 563 supports this idea. The low unblocking temperature mean direction is similar in declination to the *Witte and Kent* [1991] B remagnetization direction but is shallower in inclination by about 13°. It is possible that incomplete removal of the primary magnetic mineral in these rocks or an unresolved primary magnetization with unblocking temperatures, which overlap with those carrying the characteristic remanence of the low-temperature sites, has lead to "underprinting" of the remanence by the magnetization observed in the high-temperature sites.

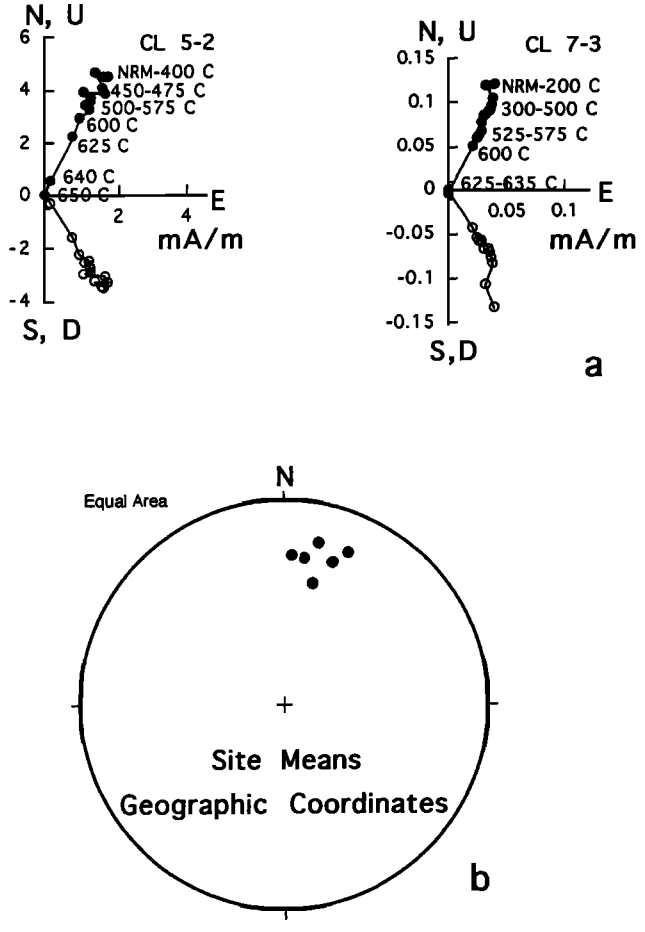


Figure 6. (a) Orthogonal vector endpoint diagrams are for thermal demagnetization of baked sediment samples from the Culpeper basin. Only high unblocking temperature behavior was observed for the Culpeper basin samples. Samples have univectorial decay into the origin to peak unblocking temperatures of 650°C. Symbols are the same as for Figure 2. (b) Equal-area stereonet shows Culpeper basin baked sediment site means in geographic coordinates. Symbols are the same as for Figure 3.

Shallow, north-northeasterly directions, similar to those reported here for the Newark basin baked sediments, have been observed in the Preakness Basalt (second Watchung flow, Newark basin) and the Holyoke Basalt (second Hartford basin flow [*Prevot and McWilliams*, 1989]). *Witte and Kent* [1990] also report shallow inclinations for the Preakness Basalt. The flows stratigraphically above and below have directions very similar to those reported for Newark basin intrusives [*Hozik*, 1992], including the Sassamansville diabase, which a baked contact test suggests is remagnetized [*Kodama and Mowery*, 1994]. *Prevot and McWilliams* [1989] propose that the middle flows in the Newark and Hartford basins record a geomagnetic field excursion near the Triassic-Jurassic boundary; however, an alternative explanation, given the evidence of a remagnetizing, hydrothermal event in the basin [*Sutter*, 1988; *Witte and Kent*, 1991], is that the middle flows in each basin were protected from the hydrothermal fluids and suffered less magnetic overprinting than the flows above and below. In fact, *Witte and Kent* [1990] report that the basalt samples they analyzed from the Newark basin extrusive zone exhibit overlapping coercivity spectra and a trend to shallower inclinations during alternating field demagnetization. These observations lead Witte and Kent to conclude that the Newark

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basin basalt directions reported in the literature are actually steeper than the true directions in these rocks.

Culpeper Basin

Only high unblocking temperature thermal demagnetization behavior was observed for the Culpeper basin baked sediments. This is consistent with thermal resetting of the hematite-bearing red beds (Bull Run Formation, Norian age [*Gore et al..*, 1989]) that were intruded by the Culpeper dikes and sills. It suggests that the remagnetization event, so prevalent in the Newark basin, either did not occur in the Culpeper basin or was of much more limited extent. *Sutter*'s [1988] geochronologic work does indicate that the low-temperature event he observed in the Newark basin and uses as evidence for a hydrothermal event also affected Culpeper basin rocks. Our observations would tend to support more limited hydrothermal activity in the Culpeper basin.

A limited remagnetization event in the Culpeper basin is consistent with the apparent passage of a baked contact test at the Goose Creek Dam sites, although the passage of this test is somewhat equivocal, as was pointed out in the Results section. A more robust result could be derived by comparing the magnetizations reported from the Culpeper basin dikes and sills [*Raymond*, 1982; *Hozik*, 1992] with the baked sediment results reported here. When the paleomagnetic poles for the sills and dikes are compared in geographic and stratigraphic coordinates, the agreement is better in geographic coordinates; however, at the 95% confidence level the dike and sill paleopoles are indistinguishable in either geographic or stratigraphic coordinates (Table 4) [*McFadden and Lowes*, 1981]. Therefore, in subsequent analyses the sill and dike paleopoles will be averaged together.

When the Culpeper basin sill and dike paleopoles are compared with the baked sediment paleopoles, they are indistinguishable in geographic coordinates but distinct in stratigraphic coordinates at the 95% confidence level (Table 4) [*McFadden and Lowes*, 1981]. This result supports our observation of a positive baked contact test at the Goose Creek Dam sites and indicates that the Culpeper basin intrusive rocks erupted into the basin's sedimentary rocks after the sediments had been tilted tectonically. This interpretation is consistent with *Smith*'s [1976] work on Boyd's sill in the Culpeper basin. The paleomagnetic pole for the Culpeper basin dikes, sills, and baked sediments, based on geographic coordinates, lies at 65.5°N, 73.1°E (A_{95}=3.7°).

Sutter's [1988] geochronological study of the intrusive igneous rocks from the Newark and Culpeper basins gives strong evidence that the eruptive episodes in these two basins are coeval. Therefore the magnetization of the baked sediments from the Newark basin, which we have demonstrated in this paper is a prefolding remanence, should be comparable with a posttilting magnetization in the dikes, sills, and baked sediments of the Culpeper basin. In fact, when the Newark basin high unblocking temperature baked sediment paleopoles, based on stratigraphic coordinates, are averaged with the Culpeper basin dikes, sills, and baked sediment paleopoles in geographic coordinates and then with the Culpeper poles in stratigraphic coordinates, the precision parameter is significantly improved (at the 95% confidence level [*McElhinny*, 1964]) when the Culpeper poles are based on geographic coordinates (Figure 7). This provides further evidence that the Culpeper sediments were baked after they had been tectonically tilted and allows us to calculate a 201 Ma paleopole for eastern North America, based on 23 sites from the Newark and Culpeper basins. This paleopole lies at 60°N, 69°E (Table 5).

Since all the sites used to calculate this paleopole are normal polarity, there is some concern that secular variation has not been adequately averaged. However, the angular dispersion for this paleopole is 12.1°, which is close to expected angular dispersion of 13°, based on the behavior of the geomagnetic field for the past 5 m.y. [*Merrill and McElhinny*, 1983]. This estimate assumes an average paleolatitude of 8° for the combined Newark and Culpeper sites. In fact, secular variation of the geomagnetic field may have been different in the Late Triassic/Early Jurassic than it has been for the past 5 m.y. Analysis of secular variation for the past 110 m.y. indicates that the angular dispersion of secular variation for a paleolatitude of 8° has varied from 19° to 12° over the past 110 m.y. [*Merrill and McElhinny*, 1983]. On the basis of the available data, it seems reasonable to assume that the sites recording the Newark/Culpeper baked sediment pole have adequately averaged secular variation and that the single polarity observed may reflect the relatively rapid emplacement of the igneous intrusions [*Olsen*, 1989]. These assumptions are supported by similar age results from the Moenave Formation [*Ekstrand and Butler*, 1989] and the Newark basin

Table 4. Culpeper Basin Dikes, Sills, and Baked Sediment Mean Paleopoles

Paleopoles	Coordinate System	Latitude, °N	Longitude, °E	*N*	*K*	γ_{c95}	γ_o
Dikes	Geographic	69.7	068.5	4	162.8	11.0	4.9
Sills	Geographic	65.6	075.7	6	64.7		
Dikes	Stratigraphic	69.5	103.2	4	78.6	7.8	7.1
Sills	Stratigraphic	65.7	087.4	6	494.1		
Dikes& Sills	Geographic	67.3	073.1	10	85.9	8.1	4.7
Baked Seds	Geographic	62.6	073.2	6	98.0		
Dikes & Sills	Stratigraphic	67.4	093.1	10	135.3	7.9	10.3
Baked Seds	Stratigraphic	67.0	119.9	6	61.2		

Dikes and sills paleopoles from *Raymond* [1982] (as published by *Hozik* [1992]; γ_{c95}, critical angle between the paleopoles [*McFadden and Lowes*, 1981] at the 95% confidence level; γ_o, observed angle between the paleopoles. Other abbreviations same as for Table 1.

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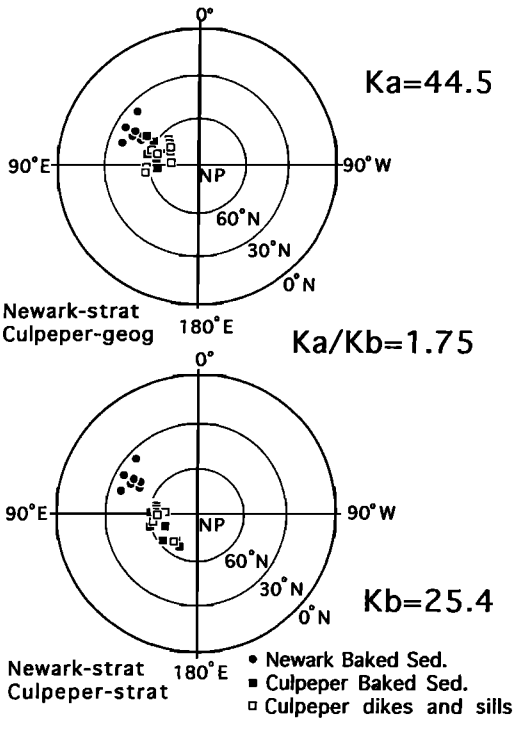


Figure 7. Paleomagnetic poles are for seven Newark basin high unblocking temperature baked sediment sites (solid circles), the six Culpeper basin baked sediment sites (solid squares), and the 10 Culpeper basin dikes and sills sites(open squares) [*Raymond*, 1982] (published by *Hozik* [1992]). In both plots the Newark basin paleopoles are based on stratigraphic coordinates, since this magnetization passes the fold test. The 15° counterclockwise rotation has been removed from the four sites located on the Sassamansville fold (see Figures 3 and 4 and text). The Culpeper basin baked sediments, sills, and dikes site means in (top) geographic and (bottom) stratigraphic coordinates are compared with the Newark results. The paleomagnetic poles show a significantly better clustering at the 95% confidence level when the Culpeper results are based on geographic coordinates [*McElhinny*, 1964]. *K*a is the precision parameter for the dispersion of paleopoles when the Culpeper basin results are based on geographic coordinates, and *K*b is the precision parameter for the dispersion of paleopoles when the Culpeper basin results are based on stratigraphic coordinates. These results suggest that the Culpeper basin sills and dikes intruded into sedimentary rocks which had already been tilted.

Hettangian red beds [*Witte and Kent*, 1990], which are either dominantly or exclusively normal polarity.

Implications for Colorado Plateau Rotation and North American Apparent Polar Wander

One of the arguments that *Kent and Witte* [1993] make for significant (>10°) clockwise rotation of the Colorado Plateau is the lack of significant apparent polar wander for North America recorded in the Late Triassic rocks of the Newark basin. They argue that since paleomagnetic poles recorded by the Newark basin sedimentary sequence from the Carnian to the Hettangian vary little from a position near 55°N, 95°E, North America did not move very much during this 25-m.y. time period. However, *Bazard and Butler* [1991] observe nearly 17° of apparent polar wander over the same period, but exclusively from rocks which lie on the Colorado Plateau. The baked sediment pole reported here (Table 5) is clearly in disagreement with the Hettangian results reported by *Witte and Kent* [1990], based on analysis of sedimentary rocks collected in the Watchung syncline near the Newark basin border fault in New Jersey. One explanation for the discrepancy is that counterclockwise rotations, like those documented for the Sassamansville syncline area [*Kodama et al.*, 1994; *Kodama and Mowery*, 1994], may have affected other sampling sites close to the border fault in the Newark basin. In fact, a paleomagnetic study of the Watchung syncline [*Van Fossen et al.*, 1986] suggests that there may be previously unrecognized structural complexities near the border fault in the northern Newark basin. *Lucas et al.* [1988]) argue that left-lateral motion may have occurred along the border fault during the opening of the Newark basin. This motion could have caused counterclockwise block rotations, like those seen at Sassamansville, at other locations along the border fault. *Kodama et al.* [1994] also point out that about 2/3 of the sites which determine the middle Newark (Norian) paleomagnetic pole reported by *Witte et al.* [1991] come from the Jacksonwald syncline, which is located along the border fault just south of the Sassamansville syncline. They suggest slight counterclockwise rotations for the Jacksonwald syncline which, if removed from the middle Newark paleopole, would result in a 7° westward displacement of the pole (from 56°N,95°E to 54°N,88°E). If the 201 Ma baked sediment pole reported here (Table 5) is combined with the Carnian results (lower Newark pole) for the Newark basin [*Witte and Kent*, 1989] and a corrected Norian pole (middle Newark) [*Witte et al.*, 1991; *Kodama et al.*, 1994], the result is nearly 19° of apparent polar wander from the Carnian to the Triassic/Jurassic boundary recorded in the Mesozoic basins of eastern North America (Figure 8) and an apparent polar wander rate of 0.89°/m.y., which is similar to the rate of 0.73°/m.y. observed by *Bazard and Butler* [1991] for equivalent age sedimentary rocks on the Colorado Plateau. The baked sediment pole reported here also provides strong evidence for the existence of the J1 cusp in North American apparent polar wander and hence strongly supports PEP analysis of North American apparent polar wander, at least for the latest Triassic/earliest Jurassic. The existence of the J1 cusp indicates that a significant change in North America's plate tectonic driving forces occurred at about 200 Ma.

The 201 Ma baked sediment pole (Table 5) is nearly

Table 5. Newark/Culpeper Basin Baked Sedimentary Rock Paleopole

	Value
Latitude, °N	59.7
Longitude, °E	68.8
N	23.0
K	44.5
A95, °	4.7

Abbreviations are same as for Table 1.

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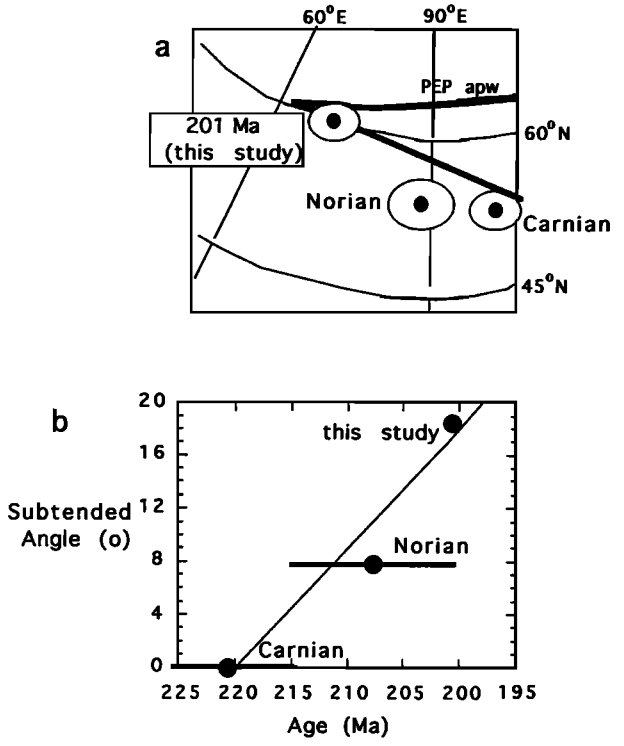


Figure 8. (a) Paleomagnetic poles from the Mesozoic basins of North America, indicating significant polar wander over the period from the Carnian to 201 Ma. The Carnian paleopole is from the Stockton, Lockatong, and Passaic Formations of the Newark basin [*Witte and Kent*, 1989]; the Norian paleopole is based on the Newark basin Passaic Formation results of *Witte et al.* [1991]; and the 201 Ma paleopole is based on the combined Newark basin baked sediments and Culpeper basin baked sediments, dikes, and sills results reported in this study. The Norian paleopole plotted here has been modified by *Kodama et al.* [1994] to remove results from the Jacksonwald syncline area, since a slight counterclockwise rotation of the area may have occurred. (b) Angular progression of paleomagnetic poles versus age is shown for the paleopoles plotted in Figure 8a. Degrees are measured in distance from the initial paleopole (Carnian paleopole). Ages for Carnian and Norian results are based on *Bazard and Butler*'s [1991] modification of the *Harland et al.* [1990] timescale. Error bars are based on the length of the stage. The best fit line to these three poles indicates a polar wander rate of 0.89°/m.y., similar to the 0.73°/m.y. observed by Bazard and Butler for paleopoles observed on the Colorado Plateau. Given the range of absolute ages assigned for the Carnian, the rate of polar wander could be as low as 0.71°/m.y., thus in agreement with Bazard and Butler's results. The Newark/Culpeper 201 Ma paleopole errors are smaller than the point plotted, based on the 1.2-1.3 m.y. analytical errors reported by *Sutter* [1988] for the crystallization age of the diabase which baked the Newark and Culpeper sediments.

equivalent in position to the Pliensbachian Kayenta Formation pole [*Bazard and Butler*, 1991] and nearly 17° from the Sinemurian/Hettangian Moenave pole (Figure 9) [*Ekstrand and Butler*, 1989]. On the basis of the *Harland et al.* [1982]) timescale, the baked sediment pole would be equivalent in age to these two poles. If the Kayenta and Moenave poles are averaged to provide a Colorado Plateau pole position equivalent in age to the Mesozoic basin 201 Ma pole, and rotation of the Colorado Plateau is assumed to have occurred around an Euler pole at 36°N,105°W [*Kent and Witte*, 1993] this would suggest about 5° of clockwise rotation for the plateau. This amount of rotation is consistent with *Bryan and Gordon*'s [1986, 1990] more detailed analysis of Colorado Plateau rotation.

Although *Kent and Witte* [1993] provide fairly strong evidence for large Colorado Plateau rotations when Norian age results are compared on and off the plateau, this result is muted somewhat by two observations. (1) Comparison of the baked sediment pole with similar age poles on the plateau only suggests small Colorado Plateau rotations. (2) Equivalent rates and magnitudes of Late Triassic apparent polar wander can now be documented on and off the plateau. Perhaps the best conclusion that can be drawn is that because of uncertainty in the exact age equivalence of units on and off the plateau, it is better to follow the approach of *Bryan and Gordon* [1986, 1990] and compare apparent polar wander over extended time periods on and off the plateau, rather than compare the results for small slices of time. This point is underscored if the more recent *Harland et al.* [1990] timescale is used for the comparison of 201 Ma poles on and off the plateau. In that case, the Moenave pole is more equivalent in age to the Mesozoic basin 201 Ma pole and larger amounts of plateau rotation could be supported.

Conclusions

A paleomagnetic study of the baked Passaic Formation sedimentary rocks in the Newark basin in southeastern Pennsylvania reveals two types of magnetic behavior. Dark

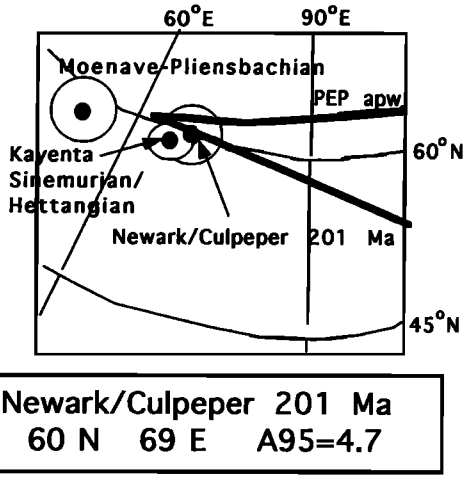


Figure 9. Comparison is shown of the Newark/Culpeper 201 Ma paleopole with similar age results from the Colorado Plateau. The close agreement between the Newark/Culpeper 201 Ma paleopole and the Kayenta and Moenave paleopoles indicates only small rotations (5°) of the Colorado Plateau are necessary. The solid line is an approximation of the paleomagnetic Euler pole apparent polar wander path for North America [*Gordon et al.*, 1984].

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gray-colored, baked sedimentary rocks have peak unblocking temperatures of 640°C, high magnetic intensities, and shallow normal polarity, northeasterly directions carried by hematite. Light gray-colored rocks have peak unblocking temperatures less than 580°C, low magnetic intensities, and intermediate inclination, normal polarity, northwesterly directions carried by magnetite. The low unblocking temperature magnetizations are secondary magnetizations, similar in declination to but shallower than the B remagnetization observed by *Witte and Kent* [1991] throughout the Newark basin. This slight discrepancy may be due to underprinting by an unresolved primary magnetization. The low unblocking temperature magnetization was probably acquired by growth of secondary magnetite during a hydrothermal event, as postulated by *Sutter* [1988], based on geochronologic data. The high unblocking temperature magnetization is significantly prefolding. Both the low unblocking temperature magnetization and the high unblocking temperature magnetization support previous evidence from sedimentary [*Kodama et al.*, 1994] and igneous [*Kodama and Mowery*, 1994] rocks for a 15° counterclockwise block rotation of the Sassamansville syncline. If this rotation is removed from the high unblocking temperature sites collected from around the fold, a stronger passage of the fold test results. Since *Sutter*'s [1988] geochronological work indicates that the intrusives in northern Virginia's Culpeper basin are the same age as those in the Newark basin, virtual geomagnetic poles (VGPs) calculated from the tilt-corrected, high unblocking temperature magnetizations isolated from the Newark basin sites were compared with the VGPs calculated from the site means of a high-temperature magnetization isolated from baked sedimentary rocks in the Culpeper basin, and they were compared with the magnetizations isolated by *Raymond* [1982] from dikes and sills. In this comparison the Culpeper poles in geographic coordinates agreed with the prefolding Newark poles significantly better than the prefolding Culpeper poles. This indicates that Culpeper intrusives erupted into already tilted sedimentary rocks. The paleomagnetic pole determined from the combined Culpeper baked sediments, dikes,and sills and the Newark basin baked sediments lies at 60°N,69°E and is of 201 Ma. age. This latest Triassic/earliest Jurassic pole, when combined with the Newark basin Carnian results [*Witte and Kent*, 1989] and Norian results [*Witte et al.*, 1991] corrected for a counterclockwise block rotation [*Kodama et al.*, 1994], provides a record of significant polar wander from eastern North America's Mesozoic basins for the Late Triassic. This is consistent with observations made for a similar time period from rocks on the Colorado Plateau (Bazard and Butler, 1991). Comparison of the Newark/Culpeper pole with similar age poles from the Kayenta [*Bazard and Butler*, 1991] and Moenave Formations [*Ekstrand and Butler*, 1989] only requires small amounts (5°) of Colorado Plateau rotation. The pole also provides the first well-dated evidence of the J1 cusp in North American apparent polar wander from rocks not located on the Colorado Plateau, thus providing strong support for the usefulness of paleomagnetic Euler pole analysis of apparent polar wander.

Acknowledgments. This work was supported, in part, by NSF grant EAR-9105879. The Culpeper basin samples were collected as part of a class project which was funded by Lehigh University.

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(Received August 9, 1993;
revised January 12, 1994;
accepted February 2, 1994.)

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